Table 13.3 Differences between Cyclic Photophosphorylation and Non-Cyclic Photophosphorylation	
Cyclic Photophosphorylation	**Non-Cyclic Photophosphorylation**
1. PS I only involved	1. PS I and PS II involved
2. Reaction centre is P700	2. Reaction centre is P680
3. Electrons released are cycled back	3. Electron released are not cycled back
4. Photolysis of water does not take place	4. Photolysis of water takes place
5. Only ATP synthesized	5. ATP and NADPH + H^+ are synthesized
6. Phosphorylation takes place at two places	6. Phosphorylation takes place at only one place
7. It does not require an external electron donor	7. Requires external electron donor like H_2O or H_2S
8. It is not sensitive to di chloro di methyl urea (DCMU)	8. It is sensitive to DCMU and inhibits electron flow

- During Non-Cyclic electron transport two NADPH + H^+ are produced and it requires 4 electrons.
- Transportation of 4 electrons requires 8 quanta of light.

Check your grasp!
Name the products produced from Non-Cyclic photophosphorylation?
Why does PS II require electrons from water?
Can you find the difference in the Pathway of electrons during PS I and PS II?

13.10.4 Chemiosmotic Theory

Chemiosmosis theory was proposed by **P. Mitchell** (1966). According to this theory electrons are transported along the membrane through PS I and PS II and connected by Cytochrome b6-f complex. The flow of electrical current is due to difference in electrochemical potential of protons across the membrane. Splitting of water molecule takes place inside the membrane. Protons or H^+ ions accumulate within the lumen of the thylakoid (H^+ increase 1000 to 2000 times). As a result, proton concentration is increased inside the thylakoid lumen. These protons move across the membrane because the primary acceptor of electron is located outside the membrane. Protons in stroma less in number and creates a proton gradient. This gradient is broken down due to the movement of proton across the membrane to the stroma through CF_O of the ATP synthase enzyme. The proton motive force created inside the lumen of thylakoid or chemical gradient of H^+ ion across the membrane stimulates ATP generation (Figure 13.15).

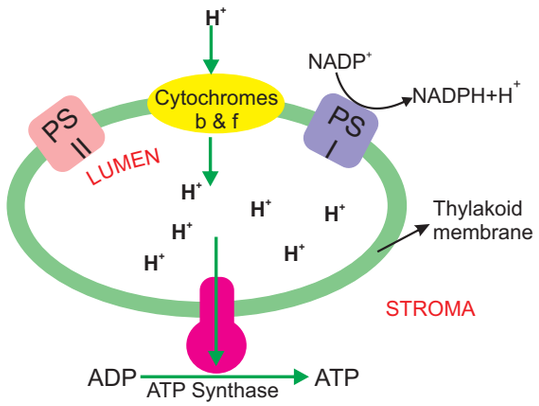


Figure 13.15: Chemiosmotic Theory

The evolution of one oxygen molecule (4 electrons required) requires 8 quanta of light. C_3 plants utilise 3 ATPs and 2 NADPH + H^+ to evolve one Oxygen molecule. To evolve 6 molecules of Oxygen 18 ATPs and 12 NADPH + H^+ are utilised. C_4 plants utilise 5 ATPs and 2 NADPH + H^+ to evolve one oxygen molecule. To evolve 6 molecules of Oxygen 30 ATPs and 12 NADPH + H^+ are utilised.

13.11 Dark Reaction or C_3 Cycle or Biosynthetic Phase or Photosynthetic Carbon Reduction (PCR)Cycle

Biosynthetic phase of photosynthesis utilises assimilatory powers(ATP and NADPH + H^+) produced during light reaction are used to fix and reduce carbon di oxide into carbohydrates. This reaction does not require light. Therefore, it is named Dark reaction. Ribulose 1,5 bisphosphate (RUBP) act as acceptor molecule of carbon di oxide and fix the CO_2 by RUBISCO enzyme. The first product of the pathway is a 3- carbon compound (Phospho Glyceric Acid) and so it is also called as C_3 Cycle. It takes place in the stroma of the chloroplast.

11th Botany_EM_Chapter 13.indd 281 06/01/2023 20:02:50